MATT KRUEGER

1. For each of the following languages, give two strings that are members and two strings that are not members – a total of 4 strings for each part. In all parts, the alphabet is $\Sigma = \{c, d\}$. (5 points)

   (a) $c(dc)^*d$

   (b) $\Sigma c \Sigma^* d \Sigma^+ d$

a) Member: cd cdcd

Not member: d dd

b) Member: cccddd dcdddd

Not member: d dd

2. For each of the following languages, give a regular expression that describes it. In all parts, the alphabet is $\Sigma = \{c, d\}$. (15 points)

   (a) $\{w \in \Sigma^* \mid w$ contains at least three c's$\}$

   (b) $\{w \in \Sigma^* \mid w$ starts with c and has odd length, or starts with d and has even length$\}$

   (c) $\{w \in \Sigma^* \mid$ the length of w is at most 5$\}$

   (d) $\{w \in \Sigma^* \mid w$ is any string except cc and ccc$\}$

   (e) $\{w \in \Sigma^* \mid$ every odd position in w is a c$\}$

a) $\Sigma^* c \Sigma^* c \Sigma^* c \Sigma^*$

b) $c(\Sigma\Sigma)^* \cup d\Sigma(\Sigma\Sigma)^*$

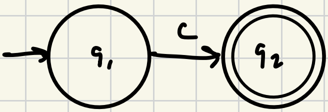
c) $\varepsilon \cup \Sigma \cup \Sigma\Sigma \cup \Sigma\Sigma\Sigma \cup \Sigma\Sigma\Sigma\Sigma \cup \Sigma\Sigma\Sigma\Sigma\Sigma$

d) $\varepsilon \cup c \cup (\Sigma\Sigma)^* d \Sigma^* \cup ccc\Sigma^+$

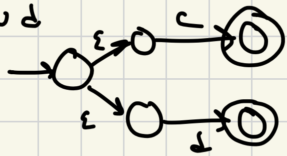
e) $(c\Sigma)^* c$

3. Convert the regular expression $(\mathtt{c} \cup \mathtt{d})^*\mathtt{ddc}^*$ into an NFA using the systematic procedure we learnt. This procedure is described in Lemma 1.55 of the text. (10 points)

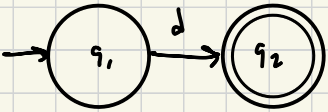
c :



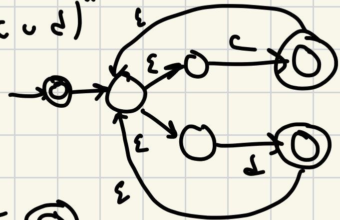
$c \cup d$



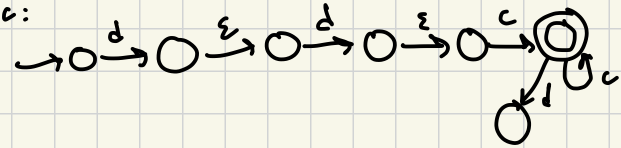
d :



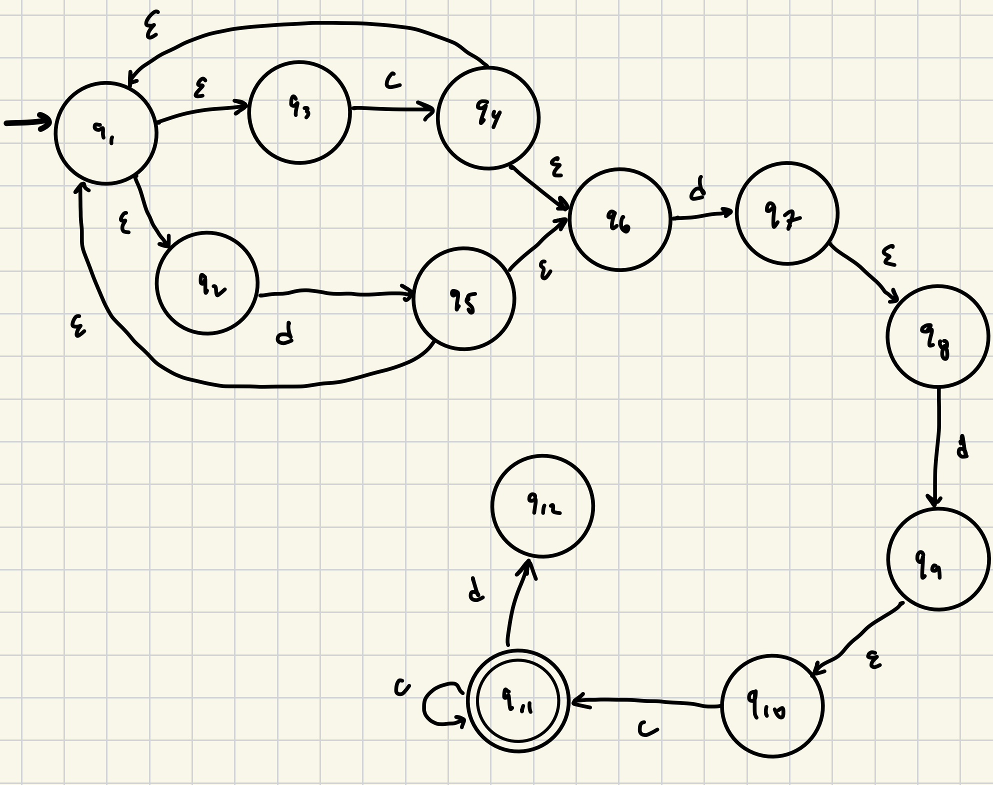
$(c \cup d)^*$



ddc :



$(c \cup d)^* ddc^*$



4. Convert the DFA in Figure 1 into a regular expression using the systematic procedure that we studied. This procedure is described in Lemma 1.60 of the text. (10 points)

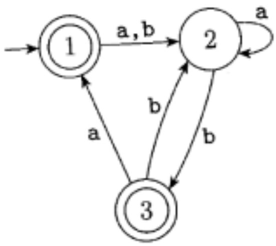
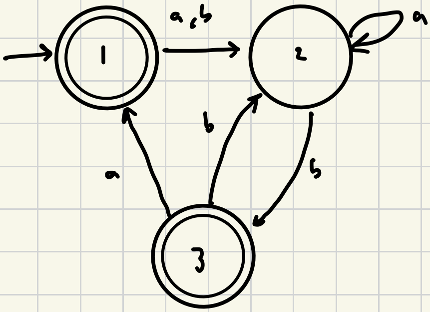
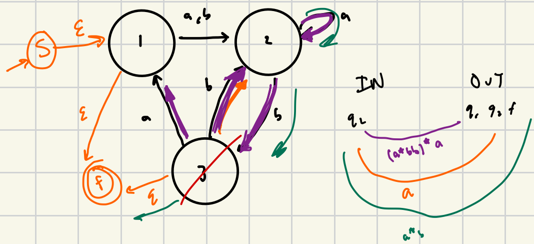
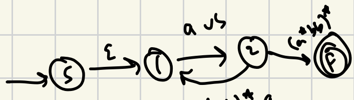
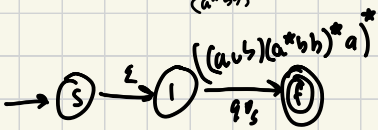


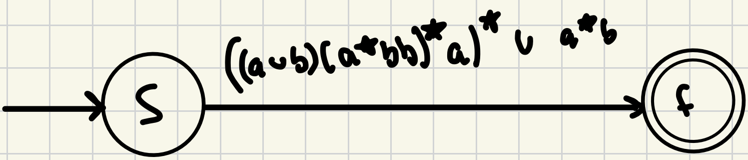
Figure 1: DFA for Problem 3











Sanity check:

$((a \cup b)(a^*bb)^*a)^*$ satisfies ① in original ✓

(a^*b) satisfies ② in original ✓

5. Show that the following languages are not regular.

(a) $\{0^n 1^m 0^n \mid m, n \geq 0\}$ (8 points)

(b) $\{w \in \{0,1\}^* \mid w$ is a palindrome$\}$. A palindrome is a string that reads the same forward and backward. (8 points)

(c) $\{w \in \{0,1\}^* \mid w$ is not a palindrome$\}$. Hint: It is easier to solve this if you don't use the pumping lemma. (4 points)

★ pumping conditions

1. for each $i \geq 0$, $xy^iz \in A$
2. $|y| > 0$, and
3. $|xy| \leq p$

a) $\{0^n 1^m 0^n\}$

Assume regular. Then there exist $p > 0$ st. any $w \in L$ with $|s| \geq p$ satisfies *pumping conditions.

choosing $w = 0^p 1^1 0^p$ as the string, the first p symbols are 0's, any y must only contain 0's from first block $\Rightarrow |xy| \leq p$

Now, pumping y, xy^iz with $i = 0$

$xy^iz = xz = 0^{p-\beta} 1^1 0^p$ $p-\beta$ leading, p trailing

this contradicts pumping lemma as β cannot be 0, thus $p-\beta \neq p \Rightarrow xy^0z \notin L$

□ not regular

b) is palindrome 010 ✓ 0110 ✓ 0011 ✗

Assume regular. Knowing palindromes are structured symmetrically, then the form of w is $0^p 1 0^p$. If regular, then it satisfies the *pumping conditions. Letting length $|x| = \alpha$ and $|y| = \beta$ $\Rightarrow 0^\alpha 0^\beta 0^{p-(\alpha+\beta)} 1 0^p$

Now, considering xy^0z $(i=0) \Rightarrow xz$

Because $|y|$ cannot be 0 $(\varepsilon) \Rightarrow xz = 0^\alpha 0^{p(\alpha-\beta)} 1 0^p$

Looking at the pumping length p,

$\Leftrightarrow \alpha + (p - (\alpha+\beta))$

$\Leftrightarrow p - \beta < p$

Thus, there are strictly less 0 if first part of string, contradicting palindrome symmetry. □ not regular

c) not palindrome

Having shown in ⓒ palindromes are NOT regular, because of the property of <u>closure under compliment</u>, $\{w \in \{0,1\}^* \mid w$ is not a palindrome$\}$ must not be regular.

□ not regular